

NAME: _____

PRACTICE MIDTERM II

Please show your work and write only in **pen**. Notes are forbidden. Calculators, and all other electronic devices, are forbidden. Brains are encouraged, but at most one may be used per exam.

1: [10 Points] Are the vectors $\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} -8 \\ -2 \\ 2 \\ 1 \end{bmatrix}$ a basis for the subspace described

by the 3-dimensional hyperplane $x_1 + 2x_2 + 3x_3 + 6x_4 = 0$? Please explain.

Let S denote the solution set of $x_1 + 2x_2 + 3x_3 + 6x_4 = 0$. Since we're already given that S is a 3-dimensional subspace, we need only verify that the vectors (a) lie in S and (b) are linearly independent. (Showing that S is a 3-dimensional subspace can be done by observing that the matrix $\begin{bmatrix} 1 & 2 & 3 & 6 \end{bmatrix}$ is already in row-echelon form, so the underlying equation has exactly 3 free variables.)

(a) can then be checked by simply plugging in the coordinates of the three vectors into the polynomial $x_1 + 2x_2 + 3x_3 + 6x_4$ and seeing that one gets 0 all three times. We omit this easy step.

(b) can be checked by Gaussian Elimination: from an earlier lecture, we know that applying Gaussian Elimination to a matrix A preserves linearly independent sets of columns.

Applying this to the matrix $\begin{bmatrix} -2 & -1 & -8 \\ 1 & -1 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$, which clearly has reduced row-echelon form

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, we see that the three vectors are clearly linearly independent. So we are done.

Note: There was a typo in the last version of this exam: the -8 above was **erroneously** written as a -4 . The **earlier** version of the problem would have thus failed the check in Part (a), changing the answer: the triple of vectors from the earlier version would thus **not** have been a basis for the stated subspace.

2: [20 Points] The following information is known about a matrix A :

$$A \begin{bmatrix} 1 \\ -2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -6 \\ 3 \end{bmatrix} \text{ and } A \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -18 \\ 9 \end{bmatrix}.$$

In fact, Ax is **always** some multiple of $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

(a) [4 Points] What is the dimension of the right-nullspace, $\mathcal{N}(A)$? Please explain why.

First note that A must be a 2×4 matrix, since A times a vector in $\mathbb{R}^4$ results in a vector in $\mathbb{R}^2$. By the 4 Subspaces Theorem, we have that the dimension of the right-nullspace plus the rank of A must be the number of columns of A . So if we know the rank, we can get the dimension of the right-nullspace. (Recall, from class, that the row space and column space of A agree, and the common value is called the rank of A .)

Now the rank of A can **not** be larger than the number of columns or rows. So the rank is at most 2. The rank can not be 0, since the statement of the problem already gives us 2 explicit nonzero vectors in the column space of A . So the rank must be 1 or 2. However, the dimension of the column space of A can't be two because the statement of the problem tells us that the column space of A is generated by a single vector. So the rank is 1 and thus the dimension of the right-nullspace is $4 - 1 = 3$.

(b) [4 Points] What is the dimension of the row space, $\mathcal{R}(A)$? Please explain why.

From our earlier observation that the row space and column space of A are equal (quoting an earlier lecture or theorem from our book), and our earlier observation that the column space has dimension 1, we see that the dimension of the row space of A is 1.

(c) [4 Points] What is the dimension of the left-nullspace, $\mathcal{N}(A^T)$? Please explain why.

The left-nullspace of A is nothing more than the right-nullspace of A^T . Observing that A^T has 2 columns, and applying the Four Subspaces Theorem again, we see that the rank of A^T plus the dimension of the right-nullspace of A^T is 2. Since $\text{rank}(A^T) = \text{rank}(A) = 1$ (the last equality following from Part (b)), we thus obtain that the dimension of the right-nullspace of A^T is 1. So the dimension of the left-nullspace of A is 1.

(d) [8 Points] Please give a solution to $Ax = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ that is **not** the zero vector.

From the first two equations given in the statement of the problem, we immediately obtain that

$$3A \begin{bmatrix} 1 \\ -2 \\ 3 \\ 1 \end{bmatrix} - A \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \end{bmatrix} = 3 \begin{bmatrix} -6 \\ 3 \end{bmatrix} - \begin{bmatrix} -18 \\ 9 \end{bmatrix}$$

Using linearity of the matrix-vector product and collecting terms, we then obtain:

$$A \left(3 \begin{bmatrix} 1 \\ -2 \\ 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Since $3 \begin{bmatrix} 1 \\ -2 \\ 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 8 \\ 1 \end{bmatrix}$ is a nonzero vector, we are done.

3: [20 Points]

What matrix A satisfies the following 3 identities?:

$$A \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, A \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix}, A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} ?$$

By block-matrix multiplication, we can clearly stack vectors next to each other and obtain the following matrix-product identity:

$$A \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 4 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 3 & -2 \\ 1 & 9 & 4 \end{bmatrix}$$

So the short answer is $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 3 & -2 \\ 1 & 9 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 4 & 1 & 1 \end{bmatrix}^{-1}$. More explicitly, via Gaussian

Elimination applied to the augmented matrix $\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 2 & -1 & 1 & 0 & 1 & 0 \\ 4 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$, one can routinely com-

pute that $\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 4 & 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} -1/3 & 0 & 1/3 \\ 1/3 & -1/2 & 1/6 \\ 1 & 1/2 & -1/2 \end{bmatrix}$ and thus $A = \begin{bmatrix} 1 & 0 & 0 \\ -2/3 & -5/2 & 7/6 \\ 20/3 & -5/2 & -1/6 \end{bmatrix}$.

Extra Credit: [20 Points]

Please find, and **factor**, the characteristic polynomial of the parametric matrix $\begin{bmatrix} a & 0 & 0 & a \\ 0 & b & c & 0 \\ 0 & c & b & 0 \\ a & 0 & 0 & a \end{bmatrix}$.

Going back to the basic definition, the characteristic polynomial is the determinant of $\begin{bmatrix} a - \lambda & 0 & 0 & a \\ 0 & b - \lambda & c & 0 \\ 0 & c & b - \lambda & 0 \\ a & 0 & 0 & a - \lambda \end{bmatrix}$. Swapping the second and fourth rows, and swapping

the second and fourth columns, we see that the last matrix has determinant equal to

that of $\begin{bmatrix} a - \lambda & a & 0 & 0 \\ a & a - \lambda & 0 & 0 \\ 0 & 0 & b - \lambda & c \\ 0 & 0 & c & b - \lambda \end{bmatrix}$. The last matrix is block-diagonal, so the deter-

minant is then just $((a - \lambda)^2 - a^2)((b - \lambda)^2 - c^2) = (\lambda^2 - 2a\lambda)(\lambda^2 - 2b\lambda + b^2 - c^2) =$

$$\boxed{\lambda(\lambda - 2a)(\lambda^2 - 2b\lambda + b^2 - c^2)}$$